

FULL SRS

Financial Institution Guarantee Management System

FIGMS — e-GP Integrated Guarantee & Securities Platform

Version: 1.0

Status: Developer Implementation Specification

Target: FPPA / e-GP / Banks / Financial Institutions / Procuring Entities / Suppliers

1. System Overview

1.1 System Name

Financial Institution Guarantee Management System (FIGMS)

1.2 Purpose

FIGMS የFinancial Institutions የሚያወጡትን የProcurement Guarantees/Securities በዲጂታል መንገድ፦

- Create
- Issue
- Approve
- Verify
- Amend
- Extend
- Cancel
- Claim
- Pay
- Release
- Audit

ለማድረግ የሚያገለግል centralized platform ነው።

2. Business Objectives

System ዋና ዓላማዎች፦

1. የPaper Guarantee ሂደትን መቀነስ
 2. Fake Guarantee ማስወገድ
 3. Guarantee verification በreal-time ማድረግ
 4. e-GP ጋር integration
 5. Bank/FI approval workflow ማቀናበር
 6. Maker–Checker–Approver control
 7. Guarantee expiry መከታተል
 8. Claim management
 9. Audit trail
 10. Central Guarantee Registry መፍጠር
-

3. Guarantee Types

System ቢያንስ እነዚህን ይደግፋል፦

3.1 Bid Security / Bid Bond

Bidder በTender ላይ ለመሳተፍ የሚያቀርበው guarantee.

3.2 Performance Guarantee

Contractor የContract ግዴታውን እንዲያከናውን የሚያስፈልግ guarantee.

3.3 Advance Payment Guarantee

Contractor የAdvance Payment ከተቀበለ በኋላ ለPE የሚሰጥ security.

3.4 Retention Guarantee

Retention money በGuarantee ለመተካት።

3.5 Custom Guarantee

ሌሎች guarantee types.

4. Actors / Users

4.1 Super Administrator

ከፖሎሊ ያለው **Role 01**።

Functions:

- Create Financial Institution
 - Activate/Deactivate Institution
 - Invite Institution Admin
 - Manage system configuration
 - Manage master data
 - Manage global permissions
 - View system-wide reports
 - Audit all institutions
-

5. Bank / Financial Institution Admin

ከፖሎሊ **Role 02**።

Functions:

- Register personnel
 - Assign roles
 - Invite users
 - Manage workflow
 - Configure branches
 - Configure guarantee products
 - Manage users
 - Manage limits
 - Manage approval hierarchy
-

6. Financial Institution Users

ከፖሎሊ **Role 03**።

User invitation:

Invitation



Accept Invitation



Email/Phone Verification



Create Password



MFA



Account Activated

Possible roles:

- Maker
- Checker
- Approver
- Guarantee Officer
- Branch Manager
- Compliance Officer
- Finance Officer
- Auditor
- Admin

7. Guarantee Management Officer

ክምስሉ **Role 04 – Manage Guarantee**

Functions:

- Receive guarantee request
- Validate applicant
- Validate tender/contract
- Enter guarantee details
- Upload documents
- Check exposure
- Prepare guarantee
- Submit to Checker
- Track status

8. Maker

Maker የGuarantee request የሚያዘጋጅ user ነው።

Maker cannot:

- Approve own request
- Change approved guarantee
- Release own guarantee
- Approve claim

Workflow:

Maker



Create Request



Submit



Checker

9. Checker

Checker የMaker ስራን ይመረምራል።

Checks:

- Customer
- Tender
- Contract
- Amount
- Expiry date
- Beneficiary
- Guarantee type
- Supporting documents
- Limits

Options:

Approve
Reject
Return for Correction

10. Approver

ክምስሉ **Role 05**::

Approver የመጨረሻ approval ይሰጣል።

System የapproval hierarchy ስልጣን መሠረት ሊያደርግ ይችላል።

Example:

Guarantee Amount	Approval
< 1M	Branch Manager
1M–10M	Regional Manager
10M–50M	Head Office
>50M	Senior Approval

እነዚህ **values configuration-based** ይሆናሉ።

11. Procuring Entity

PE የGuarantee ተቀባይ ነው።

Functions:

- View guarantee
 - Verify guarantee
 - Accept/reject
 - Request extension
 - Submit claim
 - Release guarantee
 - Download certificate
 - View status
-

12. Supplier / Contractor

Supplier:

- Select tender
 - Request guarantee
 - View guarantee status
 - Receive digital guarantee
 - Download certificate
 - Request amendment
 - Request extension
-

13. Super Admin Module

Dashboard

Display:

Total Financial Institutions
Active Institutions
Pending Institutions
Total Guarantees
Active Guarantees
Expired Guarantees
Claims
Guarantee Value

14. Financial Institution Registration

Fields:

Institution Name
Institution Type
License Number
TIN
SWIFT/BIC
Address
Phone
Email
Website
Branches
Authorized Officers
License Document
Logo
Status

Status:

Draft
Submitted
Under Review
Approved
Rejected
Suspended
Inactive

15. User Management

Admin can:

- Create user
- Invite user
- Assign role
- Assign branch
- Assign permission
- Suspend user
- Reset password
- Enable/disable MFA

16. Role-Based Access Control

RBAC structure:

Institution
↓
Department
↓
Branch
↓
User
↓
Role
↓
Permission

Example permissions:

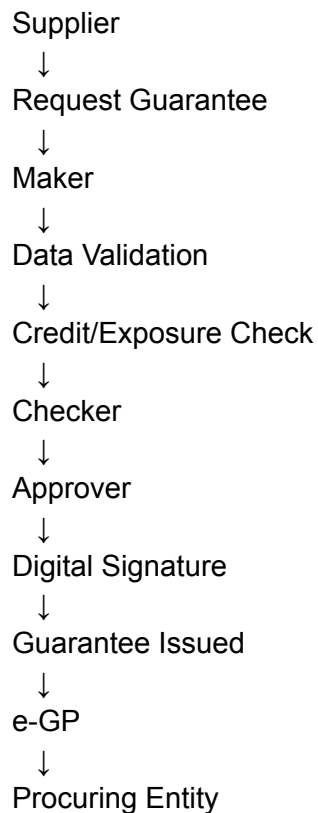
GUARANTEE_CREATE
GUARANTEE_VIEW
GUARANTEE_EDIT
GUARANTEE_SUBMIT
GUARANTEE_CHECK
GUARANTEE_APPROVE
GUARANTEE_REJECT
GUARANTEE_AMEND
GUARANTEE_EXTEND
GUARANTEE_CANCEL
GUARANTEE_RELEASE
CLAIM_CREATE
CLAIM_APPROVE
REPORT_VIEW
AUDIT_VIEW

17. Guarantee Request

Request Form

Guarantee Type
Applicant/Supplier
TIN
Tender Number
Tender Title
Contract Number
Beneficiary
Guarantee Amount
Currency
Issue Date
Effective Date
Expiry Date
Claim Expiry Date
Branch
Account Number
Purpose
Terms & Conditions
Supporting Documents

18. Guarantee Creation Workflow



19. Maker–Checker–Approver

ይህ mandatory control መሆኑን አለበት።

Maker
↓
Checker
↓
Approver
↓
Issued

Rule

Maker ≠ Checker

Checker ≠ Approver

Maker ≠ Approver

Same user በራሱ transaction ላይ approval ማድረግ **MUST NOT** ይሆናል።

20. Guarantee Number

System automatically generates unique ID.

Example:

ETH-BANK-2026-PG-00001245

Format:

Country
Institution
Year
Guarantee Type
Sequence

QR Code ይኖረዋል።

21. Digital Guarantee Certificate

Certificate contains:

Institution Logo
Guarantee Number
Guarantee Type
Applicant
Beneficiary
Tender Number
Contract Number
Amount
Issue Date
Expiry Date
Claim Period
Authorized Signatory
QR Code
Digital Signature
Verification URL

22. QR Verification

PE ወይም ማንኛውም authorized verifier:

Scan QR



Verification Portal



Guarantee Number



Status Check

Possible results:

-  Valid
-  Expiring Soon
-  Expired
-  Cancelled

- Claimed
- Suspended

Sensitive customer information ግን ለpublic verification አይታይም።

23. Guarantee Status

DRAFT
SUBMITTED
UNDER_CHECK
UNDER_APPROVAL
APPROVED
ISSUED
ACTIVE
AMENDED
EXTENDED
EXPIRING
EXPIRED
CLAIMED
PAID
RELEASED
CANCELLED
REJECTED
SUSPENDED

24. e-GP Integration

ይህ የSystem ዋና ክፍል ነው።

e-GP
↕
API Gateway
↕
Guarantee System
↕
Bank / Financial Institution

e-GP → FIGMS

የሚመጡ:

Tender ID
Tender Number
Tender Title
PE
Supplier
Bid Security Requirement
Required Amount
Expiry Requirement
Contract Number

FIGMS → e-GP

የሚላኩ:

Guarantee Number
Guarantee Type
Supplier
Issuer
Amount
Issue Date
Expiry Date
Status
Verification URL
QR
Digital Certificate

25. API Architecture

Authentication

POST /api/v1/auth/token

Create Guarantee

POST /api/v1/guarantees

Get Guarantee

GET /api/v1/guarantees/{id}

Verify

GET /api/v1/guarantees/{number}/verify

Approve

POST /api/v1/guarantees/{id}/approve

Reject

POST /api/v1/guarantees/{id}/reject

Amend

POST /api/v1/guarantees/{id}/amend

Extend

POST /api/v1/guarantees/{id}/extend

Release

POST /api/v1/guarantees/{id}/release

Claim

POST /api/v1/guarantees/{id}/claim

26. Claim Management

PE ၵᄡᄡᄡ:

Claim Request



Guarantee Validation



Claim Eligibility



Checker



Approver



Payment Decision



Financial Settlement

Claim fields:

Guarantee Number
Claim Amount
Claim Date
Reason
Supporting Documents
Beneficiary
Bank Account
Authorized Officer

27. Amendment Management

Guarantee ከIssued በኋላ:

- Amount Amendment
- Beneficiary Amendment
- Expiry Amendment
- Contract Amendment
- Address Amendment
- Other Terms

ሁሉም amendment naudit trail ውስጥ መመዝገብ አለበት።

28. Extension

Extension Request



Review



Approval



New Expiry Date



Digital Amendment



e-GP Update

Old value እንዲጠፋ አይፈቀድም።

29. Release Management

PE guarantee ሲለቀው:

Release Request



PE Authorization



Bank Verification



Approval



RELEASED

30. Expiry Management

System automatic scheduler ይኖረዋል።

Notifications:

30 Days Before

15 Days Before

7 Days Before

3 Days Before

1 Day Before

Expiry Date

Notification channels:

- Email
 - SMS
 - In-app
 - Dashboard
 - API webhook
-

31. Exposure / Limit Management

Bank የCustomer ወይም institution exposure ማየት አለበት።

Example:

Customer Limit = 50M

Existing Guarantees = 30M

Available Limit = 20M

New guarantee > Available Limit:

BLOCK

and configurable escalation.

32. Document Management

Supported:

PDF

JPG

PNG

DOCX

XLSX

Documents:

- Tender
- Contract
- Customer documents
- Board resolution
- Application
- Guarantee certificate
- Amendment
- Claim documents
- Release document

Document versioning mandatory.

33. Digital Signature

System የDigital Signature አቅም ይኖረዋል።

Maker Signature

Checker Signature

Approver Signature

Institution Seal

Timestamp

Certificate

Signature verification ሊደረግ ይገባል።

34. Audit Trail

Every action:

User

Role

Institution

IP Address

Timestamp

Action

Old Value

New Value

Transaction ID

Device

Example:

09/06/2026 14:21

User: BANK-MAKER-001

Action: CREATE GUARANTEE

Guarantee: ETH-001245

Audit logs ሊሰረዙ አይችሉም።

35. Notifications

Event-driven notification system:

Guarantee Created
Guarantee Submitted
Checker Approved
Checker Rejected
Approver Approved
Guarantee Issued
Guarantee Expiring
Guarantee Expired
Claim Submitted
Claim Approved
Claim Rejected
Guarantee Released

36. Reporting

Management Reports

- Total Guarantees
 - Total Guarantee Value
 - Active Guarantees
 - Expired
 - Released
 - Cancelled
 - Claimed
 - Paid
 - By Bank
 - By PE
 - By Supplier
 - By Region
 - By Guarantee Type
-

37. Dashboard

Super Admin

Institutions
Users
Guarantees

Guarantee Value
Claims
Expiring Guarantees
System Alerts

Bank Admin

My Institution
Pending Requests
Approved
Rejected
Active
Expiring
Claims
Exposure

PE

Received Guarantees
Valid
Expired
Claimed
Released

38. Search

Search by:

Guarantee Number
Supplier
TIN
Tender Number
Contract Number
PE
Bank
Amount
Date
Status
Guarantee Type

Advanced filtering mandatory.

39. Security Requirements

Authentication

- Username/password
- MFA
- OTP
- Session timeout
- Password policy
- Account lockout

Authorization

RBAC + permission-based access.

Encryption

TLS 1.2+
AES-256

Sensitive data encrypted at rest.

40. Fraud Prevention

System should detect:

Duplicate Guarantee

Same:

Supplier
Tender
Amount
Institution

duplicate হল alert.

Fake Number

Verification portal এর পূর্ণ guarantee:

INVALID

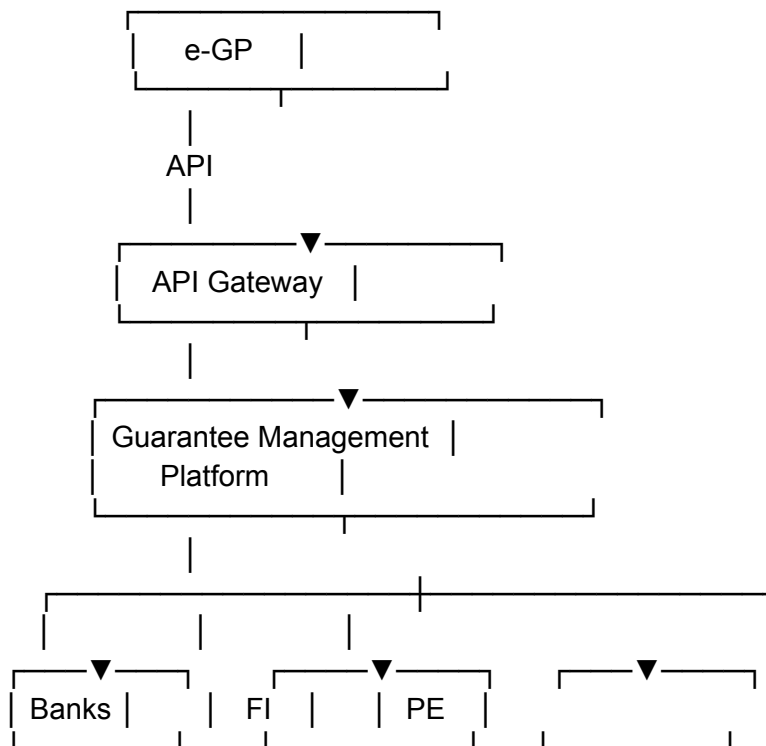
Altered Certificate

QR/Signature mismatch:

INVALID / TAMPERED

41. Integration Architecture

Recommended:



42. Recommended Backend

Developer can implement with:

Backend

Option A

Java Spring Boot
PostgreSQL
Redis
Kafka
Docker
Kubernetes

or

Option B

.NET Core
PostgreSQL
Redis
RabbitMQ/Kafka
Docker
Kubernetes

For a government/financial-grade system, I would favor **Java Spring Boot + PostgreSQL** or **.NET + PostgreSQL**.

43. Frontend

React / Next.js
TypeScript
Responsive UI
Arabic/Amharic/English-ready

Minimum languages:

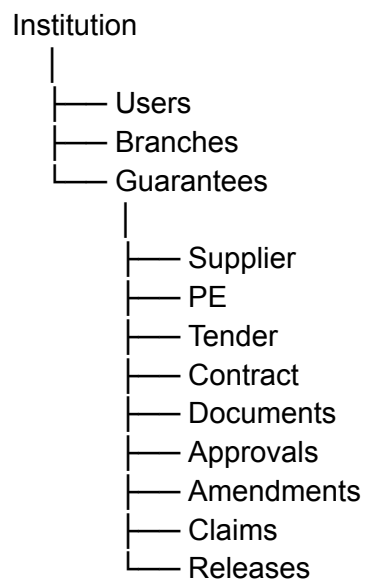
English
Amharic

44. Database Core Entities

Institution
Branch
User
Role

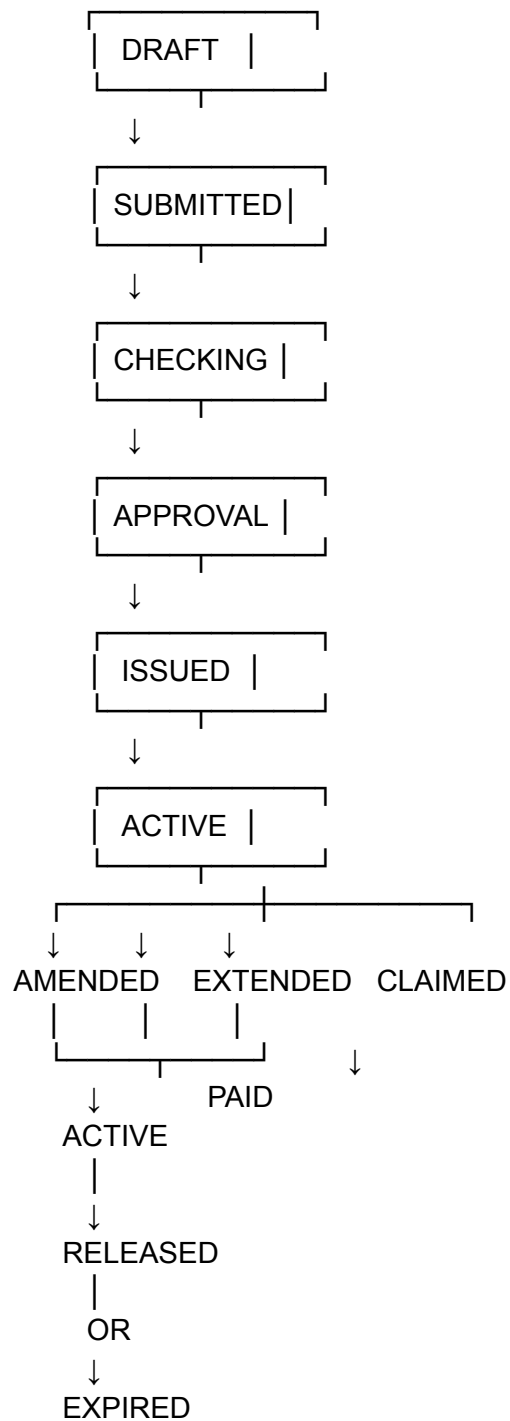
Permission
Tender
Contract
Supplier
ProcuringEntity
Guarantee
GuaranteeType
GuaranteeAmendment
GuaranteeExtension
GuaranteeClaim
GuaranteeRelease
Approval
Document
Signature
Notification
AuditLog
APIClient
Webhook
Transaction

45. Core Database Relationship



46. Guarantee Lifecycle

ይህ Developer በተለይ መከተል ያለበት core lifecycle ነው፡-



47. API Security

Every institution gets:

Client ID
Client Secret
API Key
Certificate
Webhook Secret

API authentication:

OAuth 2.0
JWT
mTLS

Financial institution ↔ e-GP communication ላይ **mTLS** መጠቀም በተለይ የሚመከር ነው።

48. Non-Functional Requirements

Availability

Target:

99.9%

Performance

Normal API:

< 2 seconds

Verification:

< 1 second target.

Scalability

System ሲገኝ:

100+ Financial Institutions
10,000+ Users
1,000,000+ Guarantees

49. Backup & Disaster Recovery

Backup

Daily Full Backup

Hourly Incremental

Point-in-Time Recovery

Disaster Recovery

Recommended:

RPO $\leq$ 15 minutes

RTO $\leq$ 1 hour

50. Monitoring

Developer የmonitoring stack ይጨምሩ፦

Application Monitoring

Database Monitoring

API Monitoring

Security Monitoring

Error Tracking

Audit Monitoring

Infrastructure Monitoring

Alerts:

API failure

Database failure

High CPU

High memory

Failed login spike

Suspicious transaction

Integration failure

51. Admin Configuration

Admin configurable:

- Guarantee types
 - Approval limits
 - Expiry rules
 - Notification rules
 - User roles
 - Permissions
 - Institution status
 - API settings
 - Document requirements
 - Workflow
 - SLA
 - Currency
 - Fee configuration
-

52. SLA Management

Example:

Maker → Checker = 4 hours

Checker → Approver = 4 hours

Approver → Issuance = 2 hours

SLA breached:

Warning



Escalation



Manager Notification

53. Fee Management

System optionally supports:

Application Fee
Issuance Fee
Amendment Fee
Extension Fee
Cancellation Fee
Claim Fee

Fee calculation:

Fixed
Percentage
Amount-based
Product-based

54. Verification Portal

Public/authorized verification page:

Enter Guarantee Number

OR

Scan QR



Verification



Issuer

Guarantee Type

Amount

Issue Date

Expiry Date

Status

Customer confidential data must not be exposed.

55. Business Rules

BR-001

Guarantee Number must be unique.

BR-002

Expired guarantee cannot be amended unless policy allows extension.

BR-003

Maker cannot approve own request.

BR-004

Cancelled guarantee cannot be reactivated without controlled process.

BR-005

Every approval must have timestamp.

BR-006

Every amendment must preserve previous version.

BR-007

Claim amount cannot exceed valid guarantee amount unless permitted by configured rules.

BR-008

Verification must always return current status.

56. Acceptance Criteria

System is accepted when:

- Bank can register.
- Bank Admin can invite users.
- Users can accept invitation.
- Maker can create guarantee.
- Checker can review.
- Approver can approve.
- Guarantee can be digitally issued.
- QR can verify it.
- PE can view/verify.

- e-GP can exchange guarantee data.
 - Amendment works.
 - Extension works.
 - Claim works.
 - Release works.
 - Expiry notification works.
 - Audit trail works.
 - Reports work.
 - Unauthorized access is blocked.
-

57. MVP — Phase 1

Developer *ಅಜ್ಞಾನುಪಮ* ರಚಿಸಿದ್ದಾರೆ

MVP

1. Institution Registration
 2. User Management
 3. RBAC
 4. Maker–Checker–Approver
 5. Guarantee Creation
 6. Guarantee Approval
 7. Digital Certificate
 8. QR Verification
 9. Guarantee Search
 10. Expiry Notification
 11. Audit Trail
 12. Basic e-GP API
 13. Dashboard
 14. Reports
-

58. Phase 2

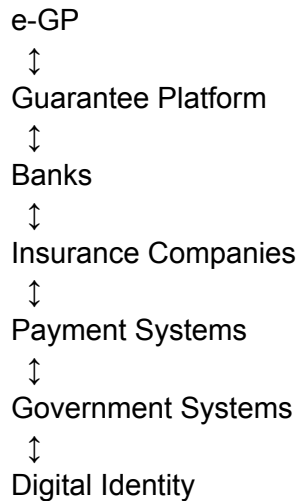
ಹಿಲ್ಸ:

- Claims
- Amendments
- Extensions

- Releases
- Digital Signature
- Advanced Exposure
- SMS
- Advanced Analytics
- Payment integration

59. Phase 3

Advanced ecosystem:



60. የDeveloper ዋና መመሪያ

Developer ላይ በቀጥታ ይህን መልእክት መስጠት ትችላለሁ፦

“Build a multi-tenant Financial Institution Guarantee Management System integrated with Ethiopia’s e-GP ecosystem. The platform must support Financial Institution registration, user/role management, Maker–Checker–Approver workflow, Bid Security, Performance Guarantee, Advance Payment Guarantee, Retention Guarantee, guarantee issuance, verification, QR-based public verification, amendment, extension, claim, release, expiry management, digital certificates, audit trails, notifications, reporting, and secure API integration with e-GP. The architecture must be enterprise-grade, scalable, secure, API-first and configurable.”

ዋናው ሀሳብ

ከምስሉ የምናየውን 5-step flow ብቻ መስራት አይበቃም። ሙሉ **Guarantee Lifecycle** መስራት አለበት፦

Registration → User → Maker → Checker → Approver → Issue → e-GP → Verify → Amend/Extend → Claim → Release/Expire → Audit.

ይህ he-GP ጋር በጣም የሚስማማ አቅጣጫ ነው፤ በተጨማሪም የኢትዮጵያ የe-GP ልማት በ2026 እየተጠናከረ መሆኑ በFPPA የቅርብ ጊዜ መረጃ ተገልጿል።

ከፈለግህ ቀጣዩን ደረጃ እሰራልሃለሁ፦ ይህን  **Full SRS** በሙሉ የ**Developer-ready PDF/Word Document** እና **Database ERD + API Specification + UI screens** ሆኖ አዘጋጅልሃለሁ።